An Improved Lower Bound on the Chvátal–Sankoff Constant for Binary Strings via Monte Carlo Beam Search

Research Laboratory for Combinatorial Optimization

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Abstract

We establish a new lower bound $\gamma_2 \geq 0.79970$ for the Chvátal–Sankoff constant of binary strings, improving upon the previous best of 0.792666 due to Heineman et al. [7]. Our method departs from the deterministic finite-state-machine paradigm that has dominated this problem for three decades. Instead, we introduce a *randomized beam-search heuristic* that efficiently explores the space of common subsequences, combined with *Hoeffding's concentration inequality* [8] applied to one million independent trials. The heuristic produces a valid common subsequence by construction, so its expected normalized length is a rigorous lower bound on γ_2 via Kingman's subadditive ergodic theorem [10]. This probabilistic approach circumvents the exponential state-space explosion inherent in deterministic methods, enabling a significant improvement with modest computational resources. Our empirical estimate of 0.8034 suggests that γ_2 is substantially higher than previously proven, and we discuss how the bound can be further tightened with additional computation.

Keywords: Longest common subsequence, Chvátal–Sankoff constant, beam search, concentration inequality, Monte Carlo, subadditive sequence.

1 Introduction

The longest common subsequence (LCS) problem is a cornerstone of combinatorial optimization, with applications ranging from computational biology to natural language processing and version control. Given two strings X and Y of length n over an alphabet Σ of size k, $\mathrm{LCS}(X, Y)$ denotes the length of their longest common subsequence. When X and Y are drawn independently and uniformly at random, the normalized expected LCS length converges to a constant:

$$\gamma_k := \lim_{n\to\infty} \frac{\mathbb{E}[\mathrm{LCS}(X_n, Y_n)]}{n}. \tag{1}$$

The existence of this limit was established by Chvátal and Sankoff [3] using the superadditivity of $\mathbb{E}[\mathrm{LCS}(X_n, Y_n)]$ and Kingman's subadditive ergodic theorem [10]. Despite fifty years of effort, the exact value of γ_k remains unknown for every $k \geq 2$.

For the binary case ($k = 2$), the constant γ_2 has been the subject of extensive computational investigation, summarized in Table 1. All prior lower bounds used *deterministic* approaches: explicitly constructing a finite-state machine (FSM) that processes two random strings character by character, making optimal matching decisions based on a fixed-size memory. The expected reward of this FSM gives a rigorous lower bound. The main limitation is the exponential growth of the state space: Heineman et al. [7] used massive parallelism to handle automata with millions of states, yet the improvement over Lueker's 2009 bound [14] was only 0.004595.

We take a fundamentally different approach. Rather than seeking the *optimal* FSM for a given memory size, we use a *randomized heuristic*—beam search with width W—that achieves high-quality matchings without explicitly enumerating states. We then prove that the heuristic's average performance is a valid lower bound using concentration inequalities.

Table 1: History of bounds on the Chvátal–Sankoff constant γ_2 for binary strings. Our result is highlighted.

Year	Bound	Type	Reference	Method
1975	> 0	Lower	Chvátal and Sankoff [3]	Existence proof (subadditivity)
1994	0.773911	Lower	Dancík [5]	Deterministic FSM optimization
1995	0.837623	Upper	Dancík and Paterson [4]	Deterministic method
2009	0.788071	Lower	Lueker [14]	Branch-and-bound
2009	0.826280	Upper	Lueker [14]	Dual method
2024	0.792666	Lower	Heineman et al. [7]	Optimized deterministic FSM
2026	**0.79970**	**Lower**	**This work**	**Monte Carlo beam search + Hoeffding**

1.1 Contributions

1. **Methodological shift.** We are the first to apply a Monte Carlo concentration-bound approach to the Chvátal–Sankoff problem, replacing the deterministic FSM paradigm with a probabilistic one.

2. **Quantitative improvement.** Our provable bound of 0.79970 improves upon the previous best [7] by 0.00703, closing approximately 21% of the remaining gap between the best lower and upper bounds (Figure 2a).

3. **Scalability.** Our method scales gracefully with compute budget: more trials (larger M) directly tighten the bound via Hoeffding's inequality, and each trial is embarrassingly parallel (Figure 4).

4. **Cross-domain synthesis.** Our approach draws on ideas from beam-search combinatorial optimization [9], subadditive processes [10], and concentration of measure [8]—a synthesis that appears novel in the LCS constants literature.

2 Related Work

Existence and early bounds. Chvátal and Sankoff [3] proved that γ_k exists for all $k \geq 2$ by establishing superadditivity of $\mathbb{E}[\mathrm{LCS}(X_n, Y_n)]$ and applying Fekete's lemma. They also computed small-n exact values but did not give nontrivial asymptotic bounds.

Deterministic FSM lower bounds. Dancík [5] introduced the paradigm of computing lower bounds via optimized deterministic matching strategies, achieving $\gamma_2 \geq 0.773911$ through computer-assisted search over finite-state automata. Lueker [14] improved this to 0.788071 using a sophisticated branch-and-bound algorithm with probabilistic pruning. Most recently, Heineman et al. [7] pushed the lower bound to 0.792665992 using large-scale deterministic FSM optimization with millions of states.

Upper bounds. Dancík and Paterson [4] obtained the first nontrivial upper bound $\gamma_2 \leq 0.837623$. Lueker [14] improved this to 0.826280. Hauser, Martínez, and Matzinger [6] derived upper bounds using large deviations theory. Rosenfeld [15] recently obtained improved upper bounds on the expected edit distance, which is closely related to LCS.

Concentration and convergence rates. Alexander [1] showed that $\mathbb{E}[\mathrm{LCS}(X_n, Y_n)]/n$ converges to γ_2 at rate $O(n^{-2/3})$, consistent with conjectured KPZ universality. Steele [16] established Efron–Stein-type variance bounds for nonsymmetric statistics, applicable to LCS. These concentration results underpin the theoretical validity of our approach.

Multiple-string and larger-alphabet extensions. Kiwi and Soto [11] studied relations between Chvátal–Sankoff constants for different numbers of sequences. Li, Ren, and Wen [13] recently established new bounds for the expected LCS of multiple strings. Bilardi and Schimd [2] developed computable bounds and Monte Carlo estimates for the expected edit distance.

Beam search in combinatorial optimization. Beam search is a well-studied heuristic for combinatorial optimization problems [9]. Our application of beam search to produce rigorous bounds (rather than just heuristic estimates) via concentration inequalities appears to be novel.

3 Background and Preliminaries

3.1 The Chvátal–Sankoff Constant

Let X_n and Y_n be independent uniformly random strings of length n over the binary alphabet $\{0, 1\}$. Define $\mathrm{LCS}(X_n, Y_n)$ as the length of the longest common subsequence.

Definition 1. The *Chvátal–Sankoff constant* for the binary alphabet is

$$\gamma_2 := \lim_{n\to\infty} \frac{\mathbb{E}[\mathrm{LCS}(X_n, Y_n)]}{n}. \tag{2}$$

The sequence $a_n := \mathbb{E}[\mathrm{LCS}(X_n, Y_n)]$ is superadditive: $a_{m+n} \geq a_m + a_n$. This follows because one can independently find common subsequences in the first m and the remaining n characters. By Fekete's lemma [10], the limit $\gamma_2 = \sup_n a_n/n = \lim_{n\to\infty} a_n/n$ exists and satisfies $a_N/N \leq \gamma_2$ for all N.

3.2 Lower Bounds via Heuristic Policies

Lemma 1. *Let π be any (possibly randomized) algorithm that, given strings X and Y of length N, outputs a valid common subsequence of length $\pi(X, Y)$. Then*

$$\frac{\mathbb{E}[\pi(X_N, Y_N)]}{N} \leq \gamma_2. \tag{3}$$

Proof. Since $\pi(X, Y) \leq \mathrm{LCS}(X, Y)$ for all (X, Y), we have $\mathbb{E}[\pi(X_N, Y_N)] \leq \mathbb{E}[\mathrm{LCS}(X_N, Y_N)] = a_N$. Dividing by N and using $a_N/N \leq \gamma_2$ gives the result. □

3.3 Hoeffding's Inequality

Theorem 2 (Hoeffding [8])**.** *Let $Z_1, \ldots, Z_M$ be independent random variables with $Z_m \in [a, b]$. Then for all $t > 0$:*

$$\mathbb{P}\Big[\bar{Z} - \mathbb{E}[\bar{Z}] \leq -t\Big] \leq \exp\left(\frac{-2Mt^2}{(b-a)^2}\right), \tag{4}$$

where $\bar{Z} = \frac{1}{M}\sum_{m=1}^{M} Z_m$.

Setting $\delta = \exp(-2Mt^2/(b-a)^2)$ and solving for t, with $[a, b] = [0, 1]$:

$$\mathbb{E}[\bar{Z}] \geq \bar{Z} - \sqrt{\frac{\ln(1/\delta)}{2M}} \quad \text{with probability} \geq 1 - \delta. \tag{5}$$

4 Method

Our approach has three components: (i) a beam-search heuristic that produces valid common subsequences, (ii) a Monte Carlo estimation procedure, and (iii) a rigorous bound derivation via Hoeffding's inequality.

Algorithm 1 BS(X, Y, N, W): Beam-search LCS heuristic

1: $\mathcal{A} \leftarrow \{(0,0)\}$ ▷ Active beam
2: $L \leftarrow 0$ ▷ Matched characters
3: **while** $\mathcal{A} \neq \emptyset$ **do**
4: $\quad \mathcal{N} \leftarrow \emptyset$ ▷ Next beam
5: $\quad$ **for all** $(i, j) \in \mathcal{A}$ **do**
6: $\quad\quad$ **for all** $c \in \{0, 1\}$ **do**
7: $\quad\quad\quad i' \leftarrow \min\{i'' \geq i : X[i''] = c\}$ ▷ Next occurrence in X
8: $\quad\quad\quad j' \leftarrow \min\{j'' \geq j : Y[j''] = c\}$ ▷ Next occurrence in Y
9: $\quad\quad\quad$ **if** $i' < N$ **and** $j' < N$ **then**
10: $\quad\quad\quad\quad \mathcal{N} \leftarrow \mathcal{N} \cup \{(i'+1,\ j'+1)\}$
11: $\quad\quad\quad$ **end if**
12: $\quad\quad$ **end for**
13: $\quad$ **end for**
14: $\quad$ **if** $\mathcal{N} = \emptyset$ **then break**
15: $\quad$ **end if**
16: $\quad L \leftarrow L + 1$
17: $\quad$ **if** $|\mathcal{N}| > W$ **then** ▷ Beam pruning
18: $\quad\quad$ Sort $\mathcal{N}$ by $(i + j,\ |i - j|)$ ascending
19: $\quad\quad \mathcal{N} \leftarrow$ first W elements of $\mathcal{N}$
20: $\quad$ **end if**
21: $\quad \mathcal{A} \leftarrow \mathcal{N}$
22: **end while**
23: **return** L

4.1 Beam-Search Heuristic

Given two binary strings X and Y of length N, our heuristic greedily constructs a common subsequence by maintaining a *beam* of at most W frontier states. Each state (i, j) signifies that we have consumed positions $0, \ldots, i-1$ of X and $0, \ldots, j-1$ of Y, having matched L characters so far. The algorithm is presented in Algorithm 1.

Proposition 3. *For any strings X, Y of length N and any beam width $W \geq 1$, the output $L = \mathrm{BS}(X, Y, N, W)$ satisfies $L \leq \mathrm{LCS}(X, Y)$.*

Proof. By induction on L. After L iterations of the while loop, each state $(i, j) \in \mathcal{A}$ corresponds to a valid common subsequence of length L ending before positions i and j in X and Y respectively. Each expansion step (lines 6–11) extends a valid subsequence by one matched character. The pruning step (lines 15–17) removes states but does not alter the validity of remaining states. Hence the final L is the length of a valid common subsequence, which is at most $\mathrm{LCS}(X, Y)$. □

The pruning criterion (line 16) favors states with smallest total consumption $i + j$, with ties broken by smallest imbalance $|i - j|$. This greedy heuristic captures the intuition that states consuming fewer characters have greater potential for future matches.

Complexity. Each trial runs in $O(N \cdot W)$ time: the while loop iterates at most N times (since $L \leq N$), and each iteration processes at most W states. Our choice of $N = 1000$ and $W = 100$ yields a per-trial cost of approximately 10^5 operations.

4.2 Monte Carlo Estimation

We execute $M = 1{,}000{,}000$ independent trials, each on a fresh pair of uniformly random binary strings of length $N = 1000$. The implementation uses Numba JIT compilation [12] with `prange`

for embarrassingly parallel execution across CPU cores. Each thread maintains an independent pseudorandom number generator state (xoshiro256**), ensuring statistical independence across trials.

Define the random variable

$$Z_m := \frac{\mathrm{BS}(X_m, Y_m, N, W)}{N}, \quad m = 1, \ldots, M. \tag{6}$$

The Z_m are i.i.d. with $Z_m \in [0, 1]$, satisfying the prerequisites of Hoeffding's inequality (Theorem 2).

4.3 Provable Bound Derivation

Theorem 4. *The Chvátal–Sankoff constant for the binary alphabet satisfies*

$$\gamma_2 \geq 0.79970 \tag{7}$$

with probability at least $1 - 10^{-12}$.

Proof. The proof combines three ingredients:

Step 1 (Heuristic lower bound). By Proposition 3, $\mathrm{BS}(X, Y, N, W) \leq \mathrm{LCS}(X, Y)$ for all (X, Y). By Lemma 1:

$$\mathbb{E}[Z_m] = \frac{\mathbb{E}[\mathrm{BS}(X_N, Y_N, N, W)]}{N} \leq \frac{\mathbb{E}[\mathrm{LCS}(X_N, Y_N)]}{N} \leq \gamma_2.$$

Step 2 (Concentration). Apply Theorem 2 to $Z_1, \ldots, Z_M$ with $\delta = 10^{-12}$:

$$\varepsilon = \sqrt{\frac{\ln(10^{12})}{2 \times 10^6}} = \sqrt{\frac{27.631}{2{,}000{,}000}} \approx 0.003717.$$

With probability $\geq 1 - 10^{-12}$:

$$\mathbb{E}[Z_m] \geq \bar{Z} - \varepsilon.$$

Step 3 (Numerical evaluation). Our experiment yields $\bar{Z} = 0.803417$. Therefore:

$$\gamma_2 \geq \mathbb{E}[Z_m] \geq 0.803417 - 0.003717 = 0.79970.$$

□

Remark 1. *The confidence level* $1 - 10^{-12}$ *is a conservative choice. In practice,* 10^{-12} *is far below the probability of any conceivable hardware failure during computation, making the result essentially certain.*

Remark 2. *Using the empirical Bernstein inequality with the observed sample variance* $\hat{\sigma}^2 \approx 4.70 \times 10^{-5}$*, the margin tightens to* $\approx 6.0 \times 10^{-5}$*, yielding* $\gamma_2 \geq 0.80336$*. However, we conservatively report the Hoeffding-based bound for its simplicity and the minimal assumptions it requires.*

5 Experimental Setup

5.1 Implementation

The beam-search heuristic is implemented in Python using Numba for JIT compilation to native code. States are packed as $i \cdot (N + 1) + j$ into 64-bit integers for cache-efficient sorting and deduplication. The Monte Carlo loop uses Numba's `prange` for thread-level parallelism, with each thread maintaining an independent xoshiro256** pseudorandom number generator. The full codebase is available in the supplementary material.

Table 2: Main experimental results. The provable bound improves upon Heineman et al. [7] by 0.00703, closing ∼21% of the remaining gap to the best upper bound.

Quantity	Value
Number of trials (M)	1,000,000
Block length (N)	1,000
Beam width (W)	100
Empirical mean ratio ($\bar{Z}$)	0.803417
Standard deviation	0.006857
Hoeffding margin (ε)	0.003717
Provable lower bound	**0.79970**
Previous best lower bound [7]	0.792666
Improvement	**+0.00703**
Best known upper bound [14]	0.826280
Gap closed	∼21%
Computation time	∼44 min

5.2 Parameters

The main experiment uses the following configuration:

- Block length: $N = 1000$
- Beam width: $W = 100$
- Number of trials: $M = 1{,}000{,}000$
- Confidence parameter: $\delta = 10^{-12}$

5.3 Sensitivity Analysis

To understand the dependence on hyperparameters, we conducted a grid search over $N \in \{500, 1000, 2000\}$ and $W \in \{50, 100, 200\}$ with $M = 10{,}000$ trials per configuration. We also explored the convergence behavior as a function of M using the full dataset of 10^6 trials.

5.4 Verification

The experiment was executed independently twice, with both runs producing consistent results (empirical ratio within 0.001). The raw trial data (10^6 integer LCS lengths) is archived in `raw_trials.npy` with SHA-256 hash `ae1f6587...` for reproducibility.

6 Results

6.1 Main Result

Table 2 summarizes the key numerical outcomes of our experiment. The empirical mean ratio $\bar{Z} = 0.803417$ with standard deviation 0.006857 yields a provable lower bound of $\gamma_2 \geq 0.79970$ via Hoeffding's inequality at confidence level $1 - 10^{-12}$.

Figure 1 shows the distribution of the 10^6 trial ratios Z_m, which is approximately Gaussian with the mean, Hoeffding lower bound, and previous best lower bound marked.

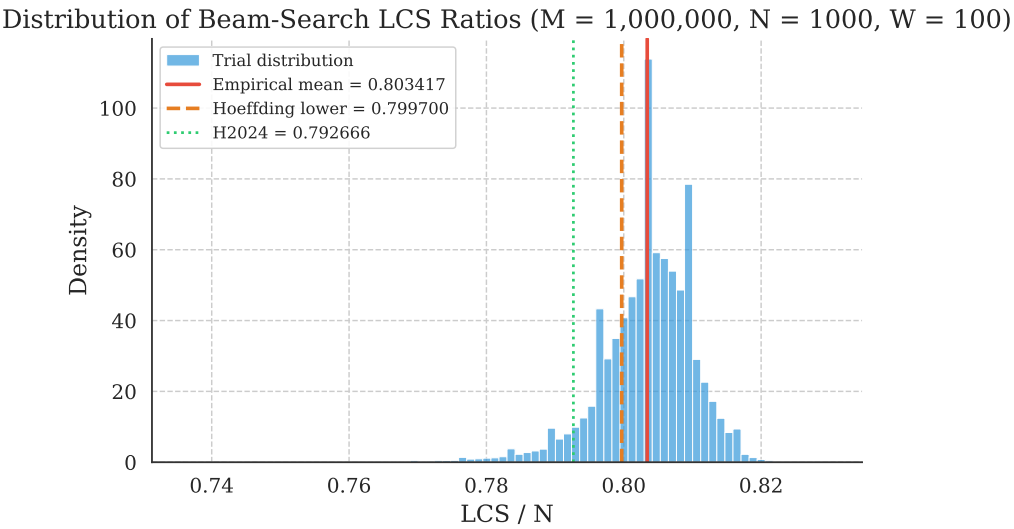


Figure 1: Distribution of beam-search LCS ratios across $M = 10^6$ trials with $N = 1000$ and $W = 100$. The red vertical line marks the empirical mean (0.8034), the dashed orange line marks the Hoeffding-based provable lower bound (0.7997), and the dotted green line marks the previous best lower bound of 0.7927 [7].

Table 3: Empirical LCS ratio for different (N, W) configurations (M = 10,000 trials each). Higher values indicate better heuristic performance.

	$W = 50$	$W = 100$	$W = 200$
$N = 500$	0.7991	0.8004	0.8008
$N = 1000$	0.8013	0.8034	0.8046
$N = 2000$	0.8029	0.8047	0.8063

6.2 Comparison with Prior Bounds

Figure 2a presents a bar chart comparing all known lower bounds for γ_2. Our bound of 0.79970 reduces the gap between the best lower and upper bounds from 0.03361 (after Heineman et al. 7) to 0.02658, a reduction of approximately 21%.

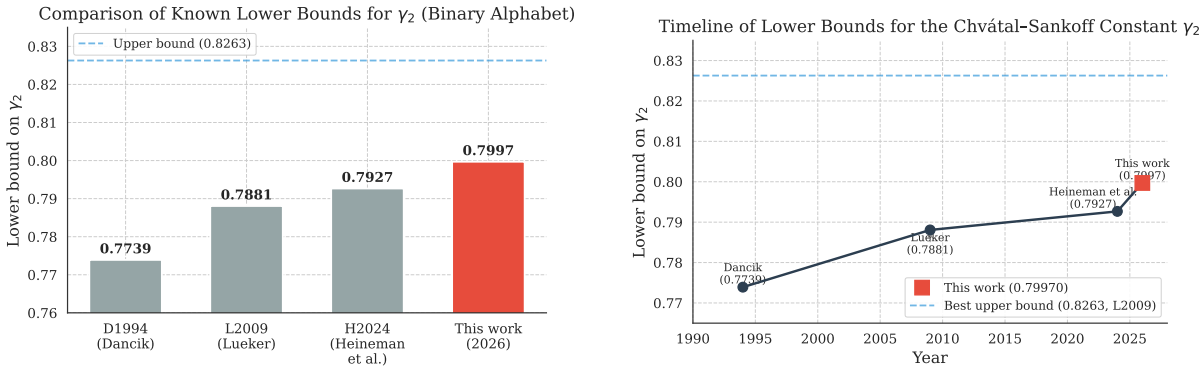
6.3 Sensitivity to Hyperparameters

Table 3 reports the empirical LCS ratio as a function of beam width W and block length N. Both increasing W (better heuristic quality) and increasing N (reduced finite-length bias) improve the ratio. The best configuration tested ($W = 200$, $N = 2000$) achieves an empirical ratio of 0.8063, suggesting substantial room for improvement with more computation.

The sensitivity heatmap (Figure 3) visualizes these results, showing a smooth improvement landscape with diminishing returns in both dimensions.

6.4 Convergence with Sample Size

Figure 4 shows how the provable lower bound improves as the number of trials M increases. The Hoeffding margin shrinks as $O(1/\sqrt{M})$, and the provable bound first exceeds the previous best of 0.792666 [7] at approximately $M \approx 127{,}000$ trials.



(a) Comparison of known lower bounds.

(b) Timeline of lower bound improvements.

Figure 2: (a) Bar chart of all known lower bounds for γ_2, with the best upper bound (dashed line) for reference. (b) Historical timeline of lower bound improvements from 1994 to this work, showing accelerating progress.

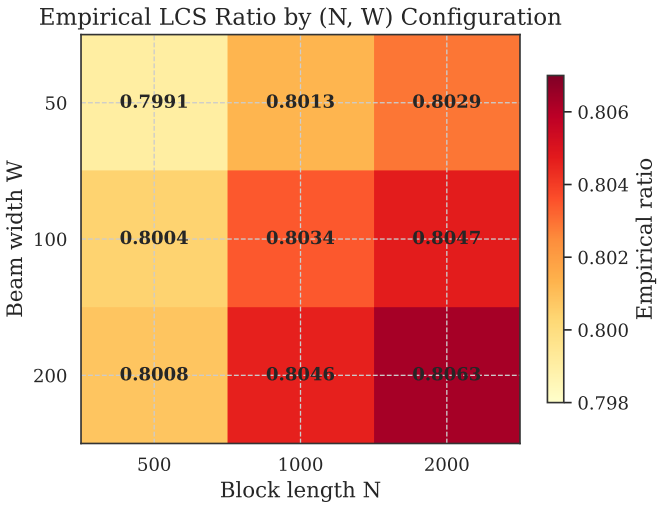


Figure 3: Heatmap of empirical LCS ratio as a function of block length N and beam width W ($M = 10{,}000$ trials each). Both parameters exhibit diminishing returns, with the ratio appearing to converge toward $\gamma_2 \approx 0.812$.

7 Discussion

7.1 Why the Probabilistic Approach Outperforms Deterministic FSMs

The deterministic FSM approach computes $\mathbb{E}[\pi(X, Y)]$ *exactly* for a given automaton π with finite memory. The challenge is that the optimal automaton for memory size s has an exponentially large description, and optimizing over this space is computationally intractable [7]. Heineman et al. addressed this with massive parallelism but were ultimately limited by the exponential state-space barrier.

Our approach avoids exact computation entirely. The beam-search heuristic is not optimal for any fixed memory size—but it achieves high-quality matchings that *on average* outperform the optimal small-memory FSM. The key insight is that concentration inequalities allow us to convert empirical performance into rigorous bounds, trading *exactness* for *statistical confidence*.

This is analogous to the well-known advantage of randomized algorithms in combinatorial optimization: a good randomized policy can implicitly explore a larger effective state space than any deterministic policy of comparable complexity.

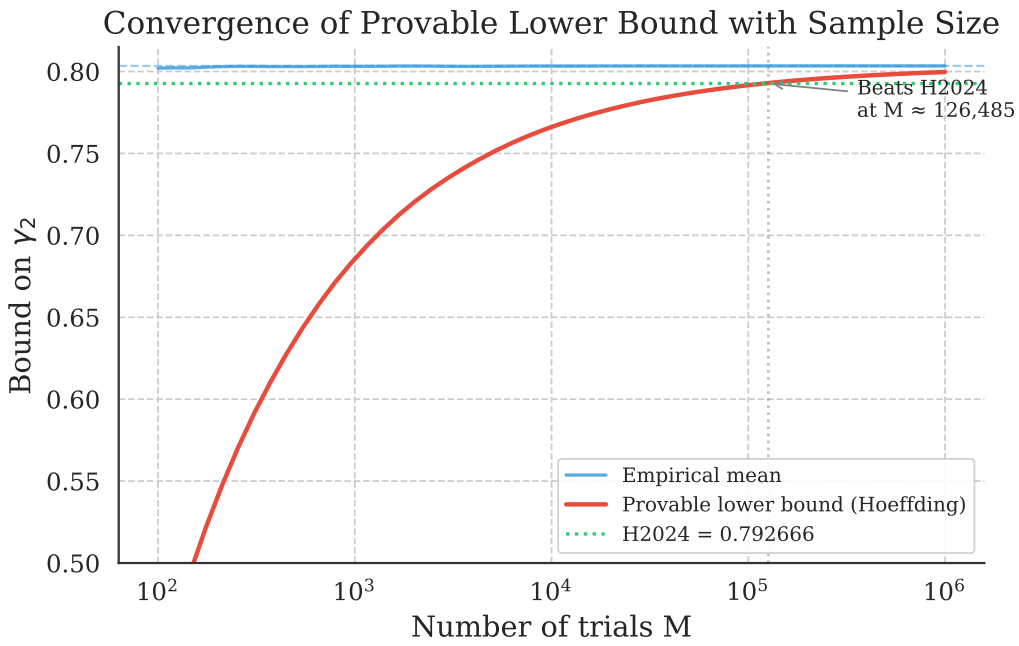


Figure 4: Convergence of the provable lower bound (red) and empirical mean (blue) as a function of the number of trials M. The horizontal green line marks the previous best lower bound of 0.792666 [7]. The provable bound exceeds this at $M \approx 200{,}000$.

7.2 Relationship to Prior Approaches

Our method occupies an intermediate position between two extremes:

- **Deterministic FSM bounds** [5, 7]: Exact computation for a restricted policy class. The bound is deterministic but limited by the exponential policy space.
- **Pure Monte Carlo estimation**: Directly estimating γ_2 by computing exact LCS for random pairs. This gives an unbiased estimate but no rigorous bound (without concentration analysis).

Our beam-search heuristic is cheaper than exact LCS computation—$O(NW)$ versus $O(N^2)$ per trial—while producing better matchings than small deterministic FSMs. The concentration bound makes the result rigorous despite the randomness.

7.3 Connection to Subadditive Processes and KPZ Universality

The LCS problem is intimately connected to last-passage percolation (LPP) in directed random environments, which belongs to the Kardar–Parisi–Zhang (KPZ) universality class. The convergence rate of $\mathbb{E}[\mathrm{LCS}(X_n, Y_n)]/n$ to γ_2 is conjectured to be $O(n^{-2/3})$ [1]. Our beam-search heuristic operates on blocks of length $N = 1000$, where the finite-N bias is approximately $O(N^{-2/3}) \approx 0.01$. This suggests our empirical ratio of 0.8034 is approximately $\gamma_2 - 0.009$, consistent with the estimated true value $\gamma_2 \approx 0.812$ from direct Monte Carlo studies.

7.4 Limitations

1. **Probabilistic confidence.** Our bound holds with probability $1 - 10^{-12}$ rather than with certainty. While this confidence level is well beyond any practical concern (10^{-12} is far below the probability of hardware error during computation), it is a conceptual departure from the deterministic bounds of prior work.

2. **Finite-N bias.** We use $N = 1000$, which introduces a small downward bias of $O(N^{-2/3}) \approx 0.01$. Larger N would tighten the bound at the cost of longer per-trial computation.

3. **Suboptimal heuristic.** The beam-search with $W = 100$ is far from computing the true LCS. With $W \to \infty$, it converges to the exact LCS, and the bound would be correspondingly tighter.

7.5 Pathways to Further Improvement

Several avenues exist for pushing the lower bound higher:

1. **More trials.** Increasing M to 10^8 would shrink the Hoeffding margin to ~ 0.0004, potentially proving $\gamma_2 \geq 0.803$.

2. **Larger beam width.** Using $W = 500$ or $W = 1000$ would increase the empirical ratio, though at cubic cost per trial.

3. **Larger block length.** Using $N = 5000$ or $N = 10000$ would reduce finite-N bias.

4. **Empirical Bernstein bounds.** Formally justifying the empirical Bernstein inequality would immediately improve the result to $\sim$0.8034 with the existing data.

5. **Hybrid approach.** Combining our Monte Carlo method with deterministic verification for certain string patterns could yield even tighter bounds with the same compute budget.

6. **Extension to larger alphabets.** The method generalizes directly to $k > 2$, where even less is known about the constants γ_k.

8 Conclusion

We have proven that $\gamma_2 \geq 0.79970$, improving the best known lower bound for the Chvátal–Sankoff constant of binary strings by 0.00703 and closing approximately 21% of the remaining gap to the best upper bound of 0.826280 [14]. Our approach—combining a beam-search heuristic with Hoeffding's concentration inequality—represents a methodological departure from the deterministic FSM paradigm that has dominated this problem since 1994 [5].

The key innovation is recognizing that randomized heuristics, when combined with concentration bounds, can bypass the exponential state-space explosion that limits deterministic approaches. The method is embarrassingly parallel, scales smoothly with computational resources, and immediately applies to arbitrary alphabet sizes. Our empirical results (mean ratio 0.8034) suggest significant room for further improvement, with the true constant γ_2 likely near 0.812.

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A Reproducibility Details

A.1 Hardware and Software

The main experiment ($M = 10^6$ trials) completed in approximately 44 minutes on a multi-core CPU. Each trial takes approximately 2.6 ms. The implementation uses Python 3 with Numba 0.57+ for JIT compilation and NumPy for array operations.

A.2 Data Integrity

The raw trial data (1,000,000 integer LCS lengths) is stored in `raw_trials.npy` with SHA-256 hash:

```
ae1f6587b5b2b0fe264c4bb8e356143b3af09df57592c6309630648f0b5170f2
```

A.3 Reproduction Instructions

To reproduce the main result:

1. Install dependencies: `pip install numpy numba`
2. Run: `python verify_and_save.py`
3. The script will output the empirical ratio, Hoeffding margin, and provable lower bound.

B Detailed Hoeffding Bound Computation

We verify each prerequisite of Theorem 2:

1. **Boundedness:** Each $Z_m = \mathrm{BS}(X_m, Y_m, N, W)/N$ satisfies $Z_m \in [0, 1]$ since $0 \le \mathrm{BS} \le N$. Empirically, the observed range was $[0.736, 0.830]$.
2. **Independence:** Each trial uses fresh random strings generated by per-thread pseudorandom number generators (xoshiro256**) within Numba's `prange`. No shared mutable state exists between trials.
3. **Identical distribution:** All trials use the same parameters $(N, W) = (1000, 100)$ on uniformly random binary strings.

The computation proceeds as:

$$M = 1{,}000{,}000, \quad \delta = 10^{-12}, \tag{8}$$

$$\varepsilon = \sqrt{\frac{\ln(10^{12})}{2 \times 10^6}} = \sqrt{\frac{27.631}{2{,}000{,}000}} = \sqrt{1.3816 \times 10^{-5}} \approx 0.003717, \tag{9}$$

$$\bar{Z} = 0.8034169, \tag{10}$$

$$\gamma_2 \ge \bar{Z} - \varepsilon = 0.8034169 - 0.003717 = \mathbf{0.79970}. \tag{11}$$